CHAPTER 6

Logarithm function

S44, S47, S49, S50

Before you begin · Exponential and inequalities

Without a calculator, in six minutes. Answer briefly. Use domains and differentiation of composite functions. Then choose the targeted review indicated under the questions.

1. Solve $e^x = 1$.

2. Give the sign of e^x.

3. Determine the domain of $\frac{1}{x-3}$.

4. Differentiate e^{2x}.

5. Solve 2 $x - 1 > 0$.

6. Solve $e^x < e^2$.

Answers

$x = 0$; strictly positive ; $\mathbb{R} \setminus \{3\}$; $2e^{2x}$; $x > \frac{1}{2}$; $x < 2$.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Recall exp is strictly increasing and to exp(0). **2.** Sketch the exponential curve above the horizontal axis. **3.** Solve x-3=0 before taking the reciprocal. **4.** Name the inner function and its derivative; R36. **5.** Isolate x by two order-preserving operations. **6.** Use the fact that exp is strictly increasing to compare the exponents. The R questions serve as follow-up material with the teacher, even if their collective routine is planned for later.

Objectives

Use the properties of ln; solve equations and thresholds; differentiate and study logarithmic functions.

Course notes

Definition and properties

The exponential is continuous, strictly increasing on $\mathbb{R}$, and its image is $]0, +\infty[$. For every $x > 0$, there is a unique real number y such that $e^y = x$: it is denoted $\ln x$.

$$\ln(e^y) = y \quad (y \in \mathbb{R}), \qquad e^{\ln x} = x \quad (x > 0), \qquad \ln 1 = 0, \quad \ln e = 1.$$

For $a, b > 0$ and $n \in \mathbb{Z}$,

$$\ln(ab) = \ln a + \ln b, \quad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \quad \ln(a^n) = n \ln a, \quad \ln\sqrt{a} = \frac{1}{2}\ln a.$$

Proof for the product. $e^{\ln a+\ln b} = e^{\ln a}e^{\ln b} = ab$; the uniqueness of the preimage gives the formula. There is no similar formula for ln $(a+b)$.

Derivative, variations and limits

We assume that ln is differentiable on $]0,+\infty[$. Differentiating $e^{\ln x} = x$ gives $e^{\ln x}(\ln x)' = 1$; so $(\ln x)' = \frac{1}{x} > 0$. Ln is strictly increasing and concave, because its second derivative is $-\frac{1}{x^2}$.

$$\lim_{x\to 0^+} \ln x = -\infty, \qquad \lim_{x\to+\infty} \ln x = +\infty.$$

Its curve is the reflection of that of the exponential in $y = x$. The line $x = 0$ is a vertical asymptote. Its tangent at 1 gives ln $x \leq x-1$ for $x > 0$.

Composition and comparative growth

If $u > 0$ is differentiable, $(\ln u)' = \frac{u'}{u}$. For every integer $m \geq 1$,

$$\lim_{x\to+\infty} \frac{\ln x}{x^m} = 0, \qquad \lim_{x\to 0^+} x^m \ln x = 0.$$

Proof at 0. Let $t = -\ln x$; when $x \to 0^+$, $t \to +\infty$ & $x\ln x = -te^{-t} \to 0$ by comparative growth. For $m \geq 1$, the same reasoning gives $-te^{-mt} \to 0$. At $+\infty$, set $t = \ln x$ to write $\frac{\ln x}{x^m} = \frac{t}{e^{mt}} \to 0$.

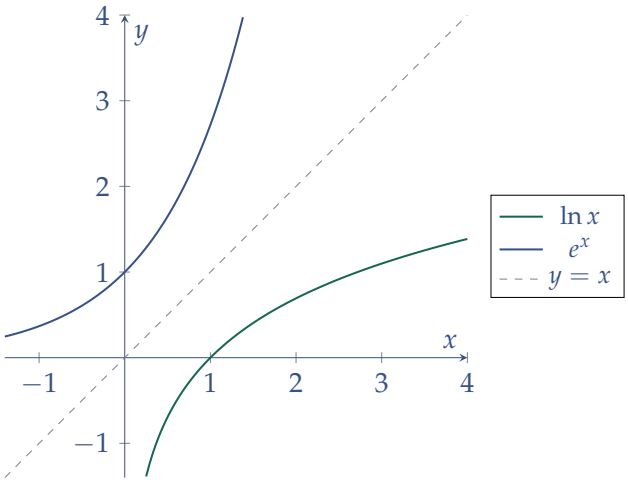
Equations and thresholds

For $A, B > 0$, $\ln A = \ln B \Leftrightarrow A = B$ and $\ln A \leq \ln B \Leftrightarrow A \leq B$. Always start with the domain.

Example. $\ln(x-1)+\ln(x+1) = \ln 8$ requires $x > 1$. The equation becomes $x^2 - 1 = 8$, so $x = 3$ only.

Bound. $0{,}8^n \leq 0.1$ is equivalent to $n\ln 0{,}8 \leq \ln 0.1$, then $n \geq \frac{\ln 0{,}1}{\ln 0{,}8} \simeq 10.319$. The first integer is 11.

Worked study. For $f(x) = \frac{\ln x}{x}$, $f' = \frac{1-\ln x}{x^2}$; f increases on $]0,e]$, decreases on $[e,+\infty[$, and its maximum is $\frac{1}{e}$. Its limit at 0 $^+$ is $-\infty$, its limit at $+\infty$ is 0.



Use the logarithm with its conditions

6.1 The inverse of the exponential

First use: S44.

Definition · Natural logarithm

For $x > 0$, $\ln x$ is the unique real number y such that $e^y = x$. Thus $e^{\ln x} = x$ for $x > 0$ and $\ln(e^y) = y$ for any real y. The logarithm transforms a positive value into a real exponent.

Worked example · Domain before transforming

To solve $\ln(x-1) = \ln(3-x)$, both arguments must be positive, so $1 < x < 3$. The fact that ln is strictly increasing then gives $x - 1 = 3 - x$, therefore $x = 2$, which belongs to the domain.

A1 · Immediate application Determine the domains of $\ln(2-x)$ and of $\ln(x^2-1)$. Solve $\ln(2x+1) = 0$, stating the domain.

6.2 Products, quotients and powers

First use: S46.

Property · Algebraic rules

For $a, b > 0$, $\ln(ab) = \ln a + \ln b$ and $\ln(a/b) = \ln a - \ln b$. For an integer m, $\ln(a^m) = m \ln a$. The conditions on the arguments must still be checked after a transformation.

Proof. Let $u = \ln a$, $v = \ln b$. Then $ab = e^u e^v = e^{u+v}$, hence $\ln(ab) = u + v$. The quotient comes from $\ln(b \times b^{-1}) = \ln 1 = 0$, and that of powers by repeating the product, then taking reciprocals. □

Worked example · An exact expression

$\ln 12 - \ln 3 = \ln 4 = 2 \ln 2$. On the other hand $\ln(1+1) = \ln 2 \neq \ln 1 + \ln 1 = 0$. The logarithm is not distributive over addition.

A2 · Two mistakes to explain Disprove " $\ln(a+b) = \ln a + \ln b$ ". Then explain why $\ln(x^2) = 2 \ln x$ is not an identity on $\mathbb{R} \setminus \{0\}$ and give the correct identity on this set.

6.3 Studying a logarithmic function

First use: S47.

Worked example · Variation and maximum

On $]0; +\infty[$, $f(x) = \frac{\ln x}{x}$ has derivative $f'(x) = \frac{1 - \ln x}{x^2}$. It increases up to $x = e$, then decreases, and its maximum is $1/e$. Its sign depends on $\ln x$, so it is negative before 1; this does not prevent it from increasing on $]0; 1[$.

A3 · Partial guidance Study the domain, derivative and sign of $g(x) = \ln(1+x^2)$. Explain why the function is even and what this implies for its variations.

6.4 The sign of $\ln q$ determines the inequality

First use: S50.

Worked example · A threshold for a decreasing sequence

For $n \in \mathbb{N}$, $0{,}8^n < 0{,}1$ is equivalent to $n \ln 0{,}8 < \ln 0{,}1$. Since $\ln 0{,}8 < 0$, dividing reverses the order: $n > \ln 0{,}1 / \ln 0{,}8 \simeq 10{,}32$. The first index is 11; $0{,}8^{10} \simeq 0{,}1074$ and $0{,}8^{11} \simeq 0{,}0859$ check the boundary. For $q > 1$, $\ln q > 0$ and the order is preserved; for $q = 1$, we do not divide by $\ln q = 0$.

A4 · Choose the right direction Determine the smallest integer $n \geq 0$ such that $1{,}05^n \geq 2$, then such that $0{,}9^n \leq 0.2$. Check the previous index in each case.

6.5 Comparing and changing variable

First use: S49.

Property · Standard limits

$\ln x \to +\infty$ at $+\infty$ & $\ln x \to -\infty$ at 0^+. For every integer $m \geq 1$, $(\ln x)/x^m \to 0$ at $+\infty$; also $x \ln x \to 0$ at 0^+. The first comparative growth result follows from $t/e^{mt} \to 0$ with $t = \ln x$.

Worked example · Adapting a comparative growth result

At $+\infty$, $\frac{2+\ln x}{x} = 2/x + \ln x / x \to 0$. At 0^+, $x^2 \ln x$ becomes $-te^{-2t}$ with t=-ln x; this expression tends to 0.

A5 · Deferred application Calculate the limits of $\ln(2x)/x$ at $+\infty$ and of $x \ln(3x)$ at 0^+. Show a transformation using algebraic rules.

A6 · Synthesis problem For each real a, determine the number of positive solutions of $\ln x = ax$. Use the study of ln x/x, its limits and the position of the level a.

A first step: Distinguish $a < 0$, $a = 0$, $0 < a < 1/e$, $a = 1/e$ and $a > 1/e$. A limit that is not attained does not give an additional solution.

One minute, spoken aloud. Explain why "taking logarithms" is not allowed on a expression whose positivity has not been verified.

Delayed review · S58 then S66. Identify a cooling constant, then find a antiderivative of $1/x$ keeping the interval of definition.

Investigation

Checking a calculation error - 30 min.

A calculation method produces an initial error $e_0 = 5$. At each step, the error is multiplied by 0.7. Let e_n be error after n steps, for $n \in \mathbb{N}$.

1. Express e_n as a function of n.

2. Using logarithms, determine the smallest integer n such that $e_n \leq 10^{-4}$.

3. Calculate, with a calculator, the error at the index found and at the previous index to check minimality of the index.

4. We now require an error at most equal to 10^{-6}. Determine the new minimum number of steps. Does dividing the tolerance by 100 multiply the number of steps by 100?

Graded exercises

E1 · Direct application. After determining the set on which it is defined, solve the equation $\ln(2x-1) = \ln(x+3)$.

E2 · Application and justification. Determine the real numbers x for which $\ln x + \ln(x-2)$ is defined, then solve $\ln x + \ln(x-2) = \ln 3$.

E3 · Method to organise. Consider $f(x) = \ln(x^2+1)$ and $g(x) = x\ln x$. State their set of definition & compute their derivatives.

E4 · Choice of strategy. Determine $\lim\limits_{x\to+\infty} \dfrac{\ln x}{\sqrt{x}}$ and $\lim\limits_{x\to 0^+} x^2 \ln x$. Justify the comparative growth results used.

E5 · Synthesis or proof. By studying a function, prove that $\ln x \leq x-1$ for every real $x > 0$. Do not use the concavity of the logarithm directly.

Work from the textbook or a past paper

Barbazo, p. 196, Exercise 36. At home after S44: deal with the three functions. Write the conditions of definition before solving the inequalities; justify the open bounds of the intervals obtained.

Homework

DM6 - Study of a logarithmic function - 55 min.

Consider the function f defined on $]0;+\infty[$ by $f(x) = \frac{1+\ln x}{x}$. Its graph is denoted $\mathcal{C}_f$.

1. Determine the limits of f at the endpoints of the domain, study its sign, then draw up its table of variations.

2. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$.

3. Prove that the equation $f(x) = \frac{1}{2}$ has exactly two solutions. For each, give a pair of bounds justified using calculated values.

4. Using the maximum of f, prove that $1 + \ln x \leq x$ for all $x > 0$. State the case of equality.

MCQ A · domain and properties

After S47; MCQ B after S50.

For each question, only one option is correct. Justify your choice orally before checking the solutions.

1. The domain of $\ln(2x-1)$ is:

A. $\mathbb{R}$ B. $[\frac{1}{2};+\infty[$

C. $]0;+\infty[$ D. $]\frac{1}{2};+\infty[$

2. For $a,b>0$, $\ln(ab)$ is:

A. $\ln a+\ln b$ B. $\ln a \ln b$

C. $\ln(a+b)$ D. $a\ln b$

3. $\ln(e^{-3})$ is:

A. 3 B. e^3

C. -3 D. $\ln 3$

4. The derivative of $\ln(1+x^2)$ is:

A. $\frac{1}{1+x^2}$ B. $\frac{2x}{1+x^2}$

C. $2\ln x$ D. $\frac{2}{x}$

5. The equation $\ln x = 2$ has solution:

A. $2e$ B. $\ln 2$

C. 2 D. e^2

MCQ B · threshold, growth and parameter

For each question, only one option is correct. Justify your choice orally before checking the solutions.

6. The smallest integer $n \geq 0$ such that $(\frac{1}{2})^n < 0.1$ is:

A. 3 B. 10

C. 4 D. 2

7. For $0 < q < 1$ and $0 < a < 1$, $q^n < a$ means:

A. $n > \frac{\ln a}{\ln q}$ B. $n < \frac{\ln a}{\ln q}$

C. $n > \frac{\ln q}{\ln a}$ D. $n > a/q$

8. $\lim_{x\to+\infty} \frac{\ln x}{x}$ is:

A. 1 B. 0

C. $+\infty$ D. $-\infty$

9. $\ln(x^2) = 2\ln x$ is valid:

A. for every real x B. for every $x \neq 0$

C. only for $x > 1$ D. for every $x > 0$

10. The maximum of $g(x) = \frac{\ln x}{x}$, $x>0$, is:

A. 1 B. e

C. $1/e$ D. 0

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. $\ln(1+1) = \ln 1 + \ln 1$.

2. Dividing an inequality by ln 2 reverses its direction.

3. $\ln x / x = 1$ has a positive solution.

MCQ solutions

1. **D.** The argument must be strictly positive: 2 $x - 1 > 0$. Zero is excluded.

2. **A.** The logarithm turns a product into a sum; no similar rule turns a sum of arguments.

3. **C.** The exponential and logarithm functions are inverses of each other on their domains.

4. **B.** Apply u'/u with $u = 1 + x^2 > 0$; the factor 2 x comes from the composite function.

5. **D.** Apply the exponential to both sides; $e^2 > 0$ satisfies the domain condition.

6. **C.** $2^{-3} = 0{,}125 \geq 0{,}1$ and $2^{-4} = 0{,}0625 < 0{,}1$. The sequence decreases, so no earlier index works.

7. **A.** Take the logarithm then divide by ln $q < 0$ reverses the direction of the inequality.

8. **B.** Comparative growth of the logarithm and a positive power of x.

9. **D.** The right-hand side requires $x > 0$. For $x \neq 0$, the general identity is ln $(x^2) = 2\ln|x|$.

10. **C.** $g' = (1 - \ln x)/x^2$, positive before e then negative. The maximum reached at e is equal to $1/e$.

True/false. False: ln 2 $\neq 0$. False: ln 2 > 0. False: the maximum 1/ $e < 1$ (question 10) rules it out.

Solutions to the applications and guidance

Application 1. Domains: $]-\infty;2[\ ;\ ]-\infty;-1[\cup]1;+\infty[$. For the equation, $x>-1/2$ and $2x+1=1$, hence $x=0$.

Application 2. Take $a=b=1$. For $x=-1$, the left-hand side $\ln 1$ exists but $\ln(-1)$ is not in the reals. The correct identity is $\ln(x^2)=2\ln|x|$ for $x\neq 0$.

Application 3. Domain $\mathbb{R}$, derivative $2x/(1+x^2)$, negative before 0 and positive after; minimum $g(0)=0$, therefore $g\geq 0$. $g(-x)=g(x)$; symmetry confirms the opposite variations on either side of 0.

Application 4. First threshold: $n\geq \ln 2/\ln 1{,}05\simeq 14{,}21$, so 15; $1{,}05^{14}\simeq 1{,}980$ and $1{,}05^{15}\simeq 2{,}079$. Second: $n\geq \ln 0{,}2/\ln 0{,}9\simeq 15.28$, so 16; $0.9^{15}\simeq 0.2059$ and $0.9^{16}\simeq 0{,}1853$.

Application 5. $\ln(2x)/x=\ln 2/x+\ln x/x\to 0$. $x\ln(3x)=x\ln 3+x\ln x\to 0$.

Application 6. $f(x)=\ln x/x$ tends to $-\infty$ at 0^+, strictly increases up to $1/e$, then decreases strictly towards 0 while remaining positive after e. Thus: one solution for $a<0$; one for $a=0$ (1) ; two for $0<a<1/e$; one for $a=1/e$ (e); none for $a>1/e$. The conclusions use the continuity on each branch.

Learning from mistakes

For the threshold, ask pupils first to predict whether the sequence increases or decreases, then to compare this direction with the inequality obtained. Have students revisit the false distributivity with a product in which one factor equals 1. The parameter problem allows work on existence, uniqueness on each branch and limits not attained.

Teaching notes and solutions

Guidance for the teacher

S44 is placed before the first mock exam, but the logarithm is not within its stated scope. Insist on the conditions $a > 0, b > 0$ before combining logarithms. The differentiability of ln is accepted, the differentiation formula is proved. For a threshold with $0 < q < 1$, the neg. sign of ln q reverses the inequality. Have the candidate index and the previous one checked.

Solutions to the activity

1. Each step multiplies the error by 0.7: $e_n = 5(0{,}7)^n$ for $n \geq 0$.

2. $e_n \leq 10^{-4}$ is equivalent to $(0{,}7)^n \leq 2 \times 10^{-5}$. Since ln is increasing, $n \ln 0{,}7 \leq \ln(2 \times 10^{-5})$. Since $\ln 0{,}7 < 0$, we get
$$n \geq \frac{\ln(2 \times 10^{-5})}{\ln 0{,}7} \simeq 30{,}33512.$$
The first suitable integer is 31.

3. $e_{30} \simeq 1{,}1270 \times 10^{-4} > 10^{-4}$ & $e_{31} \simeq 7{,}8888 \times 10^{-5} < 10^{-4}$. The sequence of errors decreases, so this check confirms minimality.

4. For the target 10^{-6}, the same calculation gives $n \geq \frac{\ln(2\times10^{-7})}{\ln 0{,}7} \simeq 43{,}2465$, so $n = 44$. The true threshold increases by $\frac{\ln 100}{|\ln 0{,}7|} \simeq 12{,}91$, then round up to the next integer. The number of steps is not multiplied by 100.

E1. We need $2x - 1 > 0$ and $x + 3 > 0$, therefore $x > \frac{1}{2}$. On this domain, the injectivity of ln gives $2x - 1 = x + 3$, i.e. $x = 4$. This value lies in the domain; the solution set is $\{4\}$.

E2. We need $x > 2$. The product formula gives $\ln(x(x-2)) = \ln 3$, so $x^2 - 2x - 3 = 0$, i.e. $(x-3)(x+1) = 0$. Only $x = 3$ satisfies $x > 2$: the other root -1 is excluded.

E3. Since $x^2 + 1 > 0$, the first function is differentiable on $\mathbb{R}$ and its derivative is $\frac{2x}{x^2+1}$. On $]0, +\infty[$, the product rule gives $(x \ln x)' = \ln x + x(\frac{1}{x}) = 1 + \ln x$.

E4. For the first limit, let $t = \sqrt{x} \to +\infty : \frac{\ln x}{\sqrt{x}} = \frac{2 \ln t}{t} \to 0$. For the second, let $t = -\ln x \to +\infty : x^2 \ln x = -te^{-2t} \to 0$. Both limits use comparative growth.

E5. For $x > 0$, let $g(x) = x - 1 - \ln x$. We have $g'(x) = \frac{x-1}{x}$, negative before 1, positive after. Its minimum is $g(1) = 0$, therefore $g(x) \geq 0$, which is equivalent to $\ln x \leq x - 1$.

Solutions to homework DM6

1. At 0^+, the numerator tends to $-\infty$ and the denominator to $0^+ : f \to -\infty$. At $+\infty$, $f = \frac{1}{x} + \frac{\ln x}{x} \to 0$. The sign is that of $1 + \ln x$: negative on $]0, e^{-1}[$, zero at e^{-1}, positive afterwards. As a quotient,
$$f'(x) = \frac{1 - (1 + \ln x)}{x^2} = -\frac{\ln x}{x^2}.$$
Thus f strictly increases up to 1, then strictly decreases; its maximum is $f(1) = 1$.

2. Differentiating $-(\ln x)x^{-2}$, we get $f''(x) = \frac{2 \ln x - 1}{x^3}$. The numerator changes sign, from negative to positive, at $x = \sqrt{e}$. Hence f is concave on $]0, \sqrt{e}]$, convex on $[\sqrt{e}, +\infty[$ and has the inflection $(\sqrt{e}, \frac{3}{2\sqrt{e}})$.

3. On $]0,1]$, continuity, strict increase, limit $-\infty$ and final value 1 give exactly one preimage of $\frac{1}{2}$. On $[1,+\infty[$, strict decrease from 1 to 0 gives exactly a second preimage. The first is in $]e^{-1},\frac{1}{2}[$, since $f(e^{-1})=0$ and $f(\frac{1}{2})=2(1-\ln 2)\simeq 0{,}6137>\frac{1}{2}$. The second is in $]5,6[$, since $f(5)\simeq 0{,}5219>\frac{1}{2}>f(6)\simeq 0{,}4653$.

4. The maximum gives $\frac{1+\ln x}{x}\leq 1$. Since $x>0$, we can multiply without changing the direction: $1+\ln x\leq x$. Equality corresponds to the unique maximum of f, hence to $x=1$.

Solutions to the textbook or past-paper work

The logarithm requires a strictly positive argument.

1. $3x-7>0\iff x>\frac{7}{3}$, therefore $D_f=]\frac{7}{3};+\infty[$.

2. $-x^2+4x-3=-(x-1)(x-3)>0$ exactly when $1<x<3$. Thus $D_g=]1;3[$.

3. We need both $x>0$ and $2-x>0$; their intersection gives $D_h=]0;2[$. The coefficient -3 placed in front of the second logarithm does not change its domain of definition.